

Milestone Project 4

Apollo Hospitals

Appointment No-Show and Patient Engagement Analysis Insights Report and Charts

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Data Analysis Period: January 2022–December 2024

Tools Used: Python, Pandas, NumPy, Matplotlib and Seaborn

Project Objective

Analyse appointment patterns, no-show behaviour, patient engagement, financial performance and service quality to identify practical opportunities to improve attendance and service delivery.

Dataset Overview

The analysis uses 75,000 appointment records and 320 doctor records, joined using `doctor_id`.

Methodology

No-show rates exclude Scheduled appointments and use Completed, No-Show and Cancelled appointments as the denominator. Realized revenue and quality metrics are analysed for Completed appointments only. Findings describe associations rather than proven causes.

Insights Reports

1. Business Overview

Monthly Appointment Volume

January 2024 recorded the highest appointment volume (2,427), while August 2023 recorded the lowest (1,785). Recurring peaks around December–January and quieter mid-year periods suggest seasonal demand.

Quarterly Appointment Volume

Q4 was the busiest quarter in every year. The highest volume occurred in 2023 Q4 (6,998 appointments), while 2023 Q3 had the lowest (5,707). This pattern can support capacity and staffing planning.

Appointment Outcomes

Completed appointments accounted for 73.70% of bookings, followed by No-Show (15.45%), Cancelled (7.72%) and Scheduled (3.13%). The no-show rate excluding Scheduled appointments was **15.94%**.

Booking Channel Mix

Apollo App contributed 45.39% of bookings and Website contributed 24.73%. Together, these channels generated 70.12% of appointments, making them important channels for patient communication.

2. No-Show Analysis

Specialty

Psychiatry had the highest no-show rate (24.52%), followed by Dermatology (21.25%). Neurology had the lowest (10.69%). General Physician contributed the most no-shows in absolute terms (3,049), so intervention priorities should consider both rates and volumes.

City

Lucknow had the highest no-show rate (17.19%), while Pune had the lowest (13.30%). Differences in patient mix, specialties and booking channels should be examined before attributing this gap to location.

Booking Lead Time

Bookings made 1–3 days ahead had the lowest no-show rate (14.33%). The rate reached 20.03% for bookings made 15–30 days ahead, supporting trials of additional reconfirmation for longer-lead bookings.

Time of Day

Evening appointments had the highest no-show rate (18.37%), followed by Afternoon (17.75%) and Morning (13.73%). Later appointments may benefit from targeted reminders and easier rescheduling.

Weekday versus Weekend

Weekend appointments had a higher no-show rate (18.68%) than weekday appointments (14.84%), a gap of 3.84 percentage points.

Appointment Type

Home Visits had the highest no-show rate (26.25%), followed by Video Consults (21.06%) and In-Clinic consultations (12.18%). Additional confirmation and clear joining instructions are useful interventions to test.

Booking Channel Risk

Walk-In had the highest recorded no-show rate (20.88%), while Apollo App had the lowest (14.18%). The definition and recording of Walk-In no-shows should be checked before operational action.

3. Reminder and Engagement Effectiveness

Reminder Type

The no-show rate was 30.16% without reminders, compared with 11.41% for SMS, WhatsApp and call reminders combined. This is an observed difference of 18.75 percentage points; it does not establish that reminders caused the reduction.

Previous No-Show History

No-show rates increased from 15.15% for patients with no previous no-shows to 32.68% for those with three or more. Prior history can help prioritise reconfirmation, although the highest-history group contained only 153 appointments.

Apollo Membership

Members had a lower no-show rate (12.66%) than non-members (16.89%). Membership was associated with better attendance, with a difference of 4.23 percentage points.

First-Time versus Returning Patients

First-time patients had a recorded no-show rate of 66.78%, compared with 8.59% for returning patients. This unusually large difference warrants validating how first-time status is recorded before using it operationally.

4. Patient and Demographic Segmentation

Age Group

Adults aged 31–45 had the highest no-show rate (18.05%), closely followed by those aged 19–30 (17.94%). Children had the lowest rate (13.54%).

Gender

Female and male patients had similar no-show rates: 16.04% and 15.84%. The small difference suggests gender alone provides little practical separation of risk.

Chronic Condition Status

Patients with chronic conditions had a no-show rate of 15.98%, compared with 15.89% for those without. Chronic condition status alone showed little difference in attendance.

Visit Reasons

Follow-up was the most common visit reason (8,807 appointments). Skin Issue had the highest no-show rate (16.79%), while Child Health had the lowest (14.64%). The overall variation across reasons was relatively small.

Age Distribution by Specialty

Paediatrics was a median age of 9 and an age range of 1–18. Gynaecology and Psychiatry both were a median age of 42 and recorded ranges of 1–85. Very young ages in these specialties warrant checking source records. These distributions count appointments, so repeat patients may appear multiple times.

5. Financial Performance

Potential Revenue at Risk

The 11,584 no-show appointments represented approximately **₹1.97 crore at listed consultation fees**. This is potential revenue at risk, not confirmed net revenue lost; it excludes replacement bookings, discounts, insurance adjustments and later rescheduling.

Average Revenue by Specialty and Appointment Type

Psychiatry generated the highest average realized revenue per completed appointment (₹2,640.45). Home Visits generated the highest average among appointment types (₹1,960.05). Higher revenue does not necessarily imply higher profit.

City Revenue

Bengaluru generated the highest realized revenue (₹99.91 lakh), closely followed by Delhi (₹99.58 lakh). City totals reflect both appointment volume and revenue per appointment.

Payment Mode Mix

Insurance was the largest recorded payment mode, accounting for 21.06% of completed appointments. Other payment modes each accounted for approximately 13%. These percentages represent appointment counts, not revenue shares.

Insurance and Out-of-Pocket Payments

Insured patients paid an average of ₹859.18 out of pocket, compared with ₹1,562.82 for uninsured patients—approximately 45% less. Average fees were similar across the two groups.

6. Doctor Utilisation and Service Quality

Doctor Utilisation

Gastroenterology had the highest average recorded utilisation (77.26%), while Psychiatry had the lowest (76.18%). The small gap suggests similar utilisation across specialties. These are appointment-weighted averages of the recorded metric.

Patient Waiting Time

Average waiting time was 11.55 minutes. Ophthalmology had the longest average wait (12.02 minutes), while Paediatrics had the shortest (11.18 minutes). Average waits varied little by time of day.

Consultation Duration and Satisfaction

The correlation between consultation duration and satisfaction was -0.0035 , indicating almost no linear relationship. Longer consultations were not consistently associated with higher satisfaction.

Doctor Experience and Fees

Doctor experience and listed fees had a modest positive correlation of 0.2780. More experienced doctors tended to charge higher fees, but specialty and city may also influence pricing.

Final Recommendations

- i. Test additional reminders for longer-lead bookings, weekend appointments and patients with previous no-shows.
- ii. Prioritise high-risk specialties alongside specialties with large missed-appointment volumes.
- iii. Validate first-time status and Walk-In no-show recording.
- iv. Make cancellation and rescheduling easier to help refill unused slots.
- v. Investigate unusually long waits and measure intervention results through controlled pilots.

Charts and Visual Analysis

1. Business Overview:

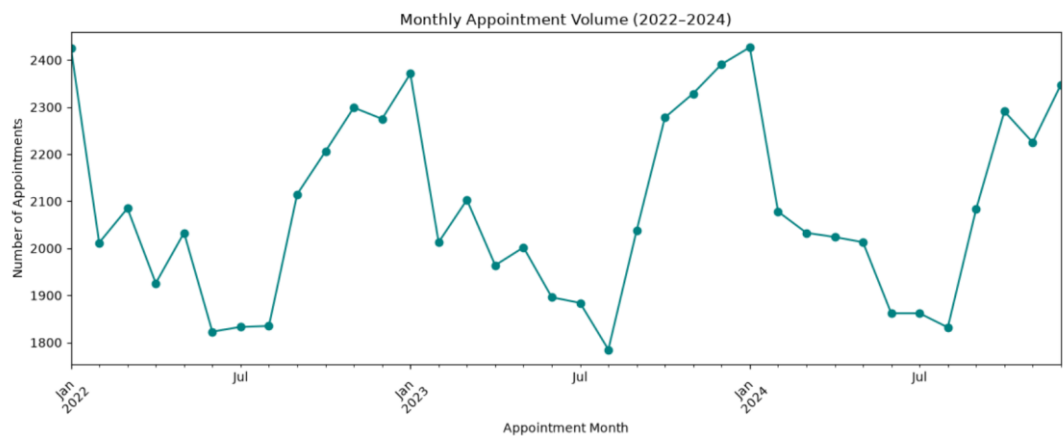


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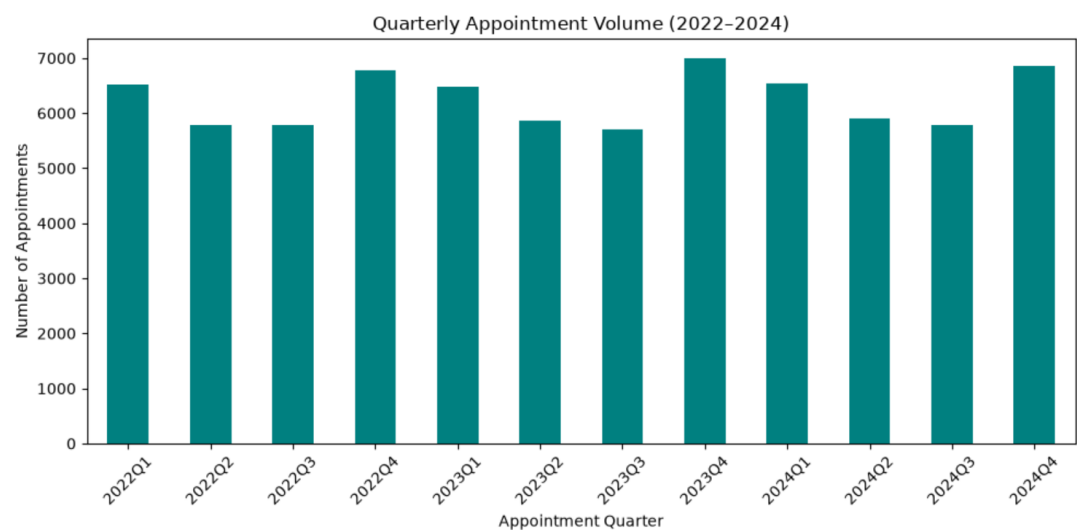


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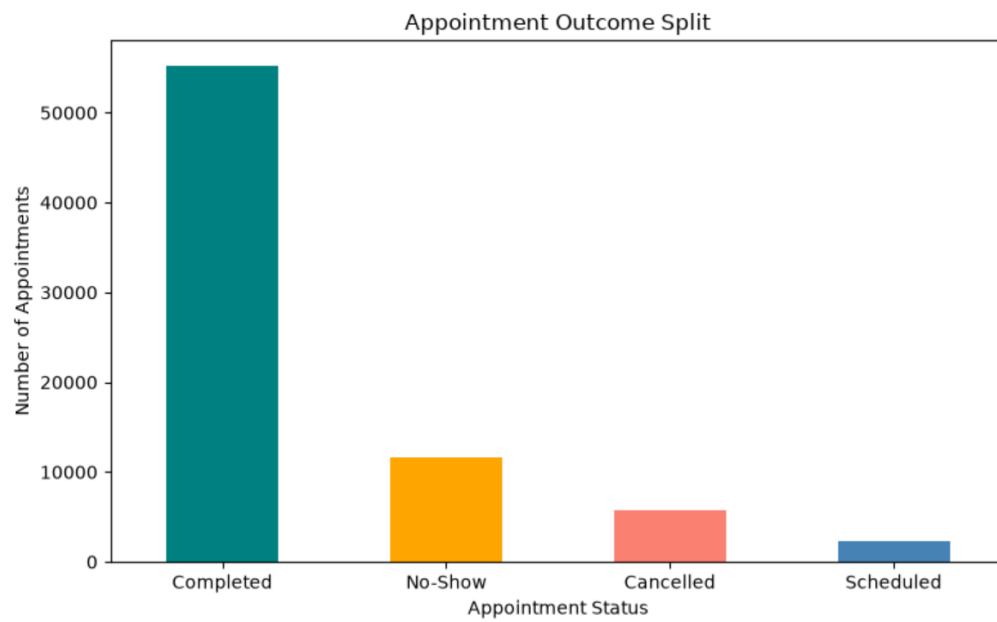


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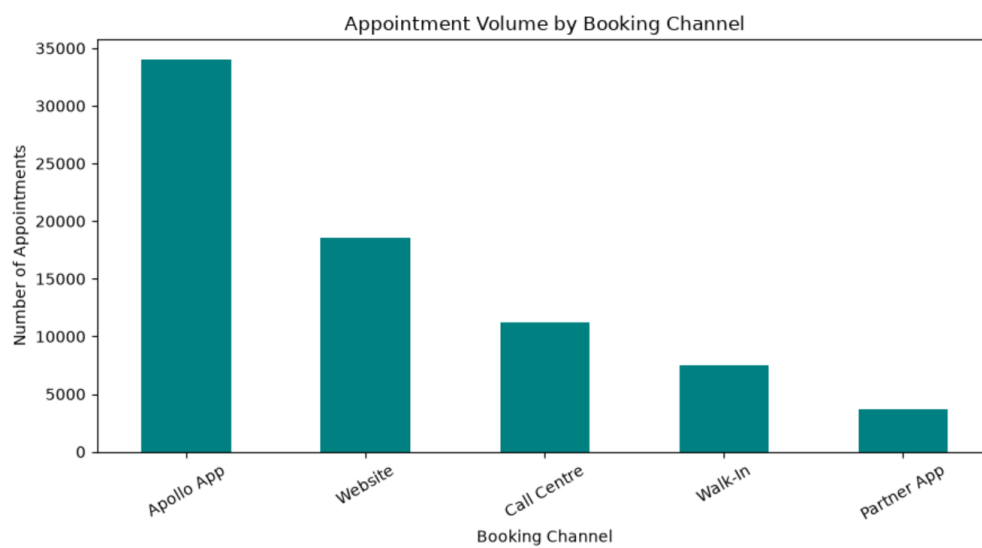


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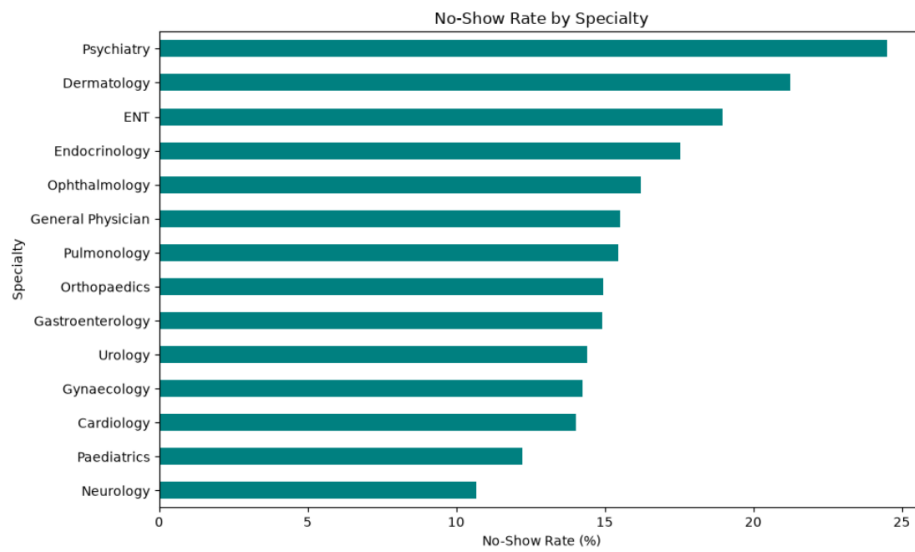


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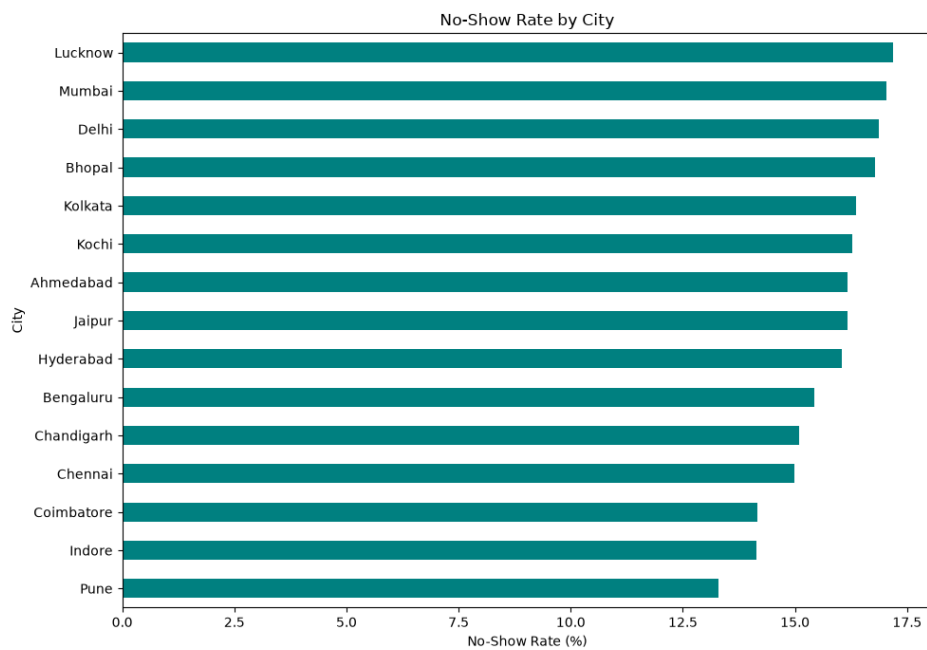


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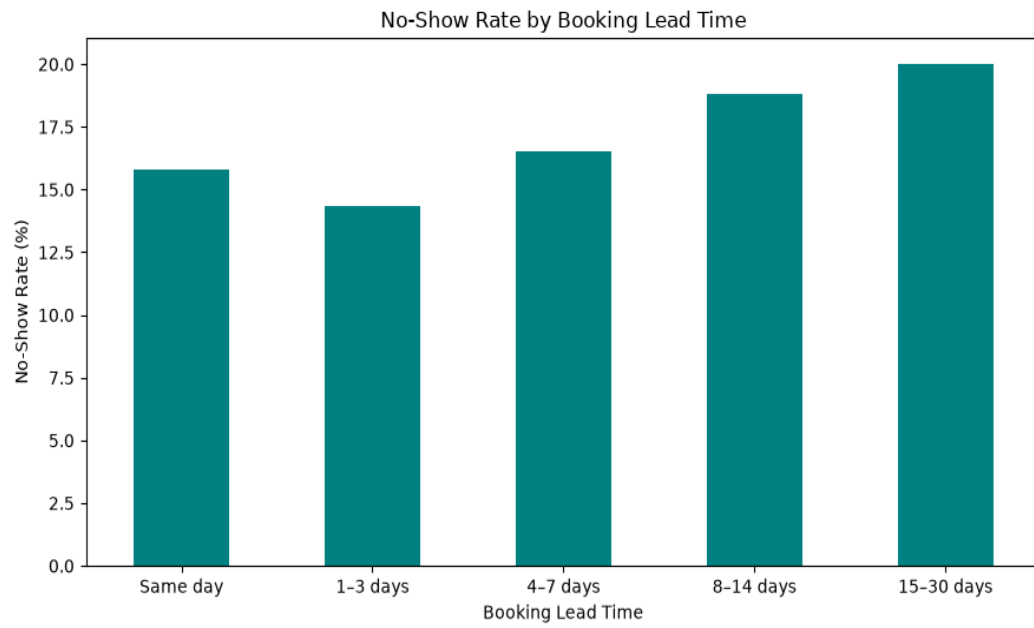


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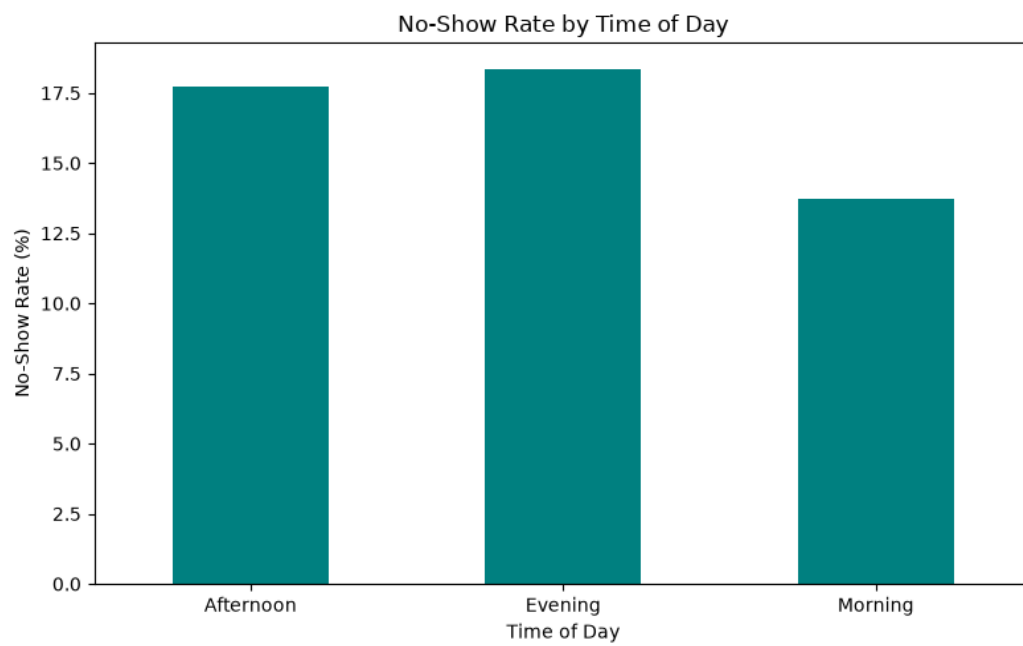


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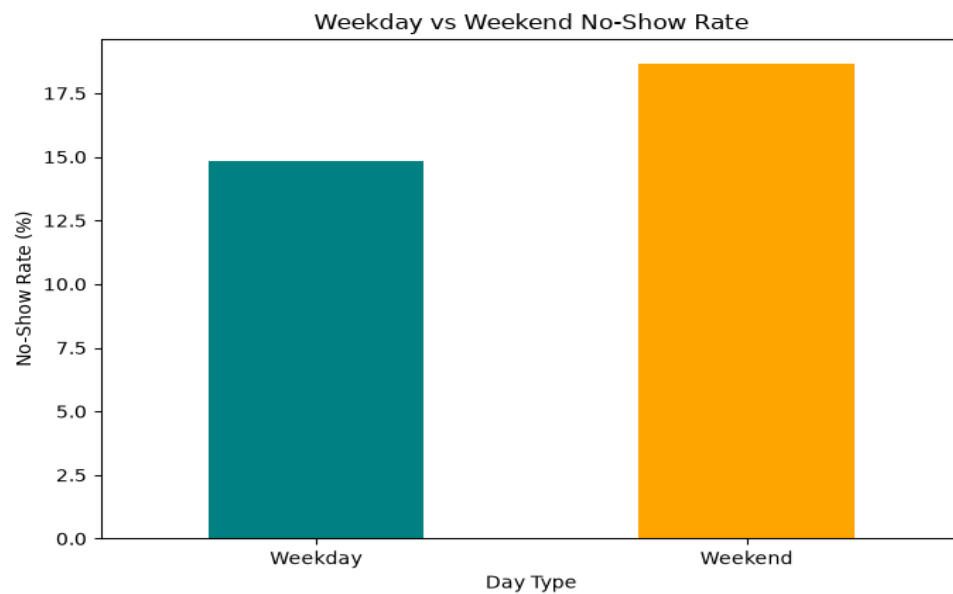


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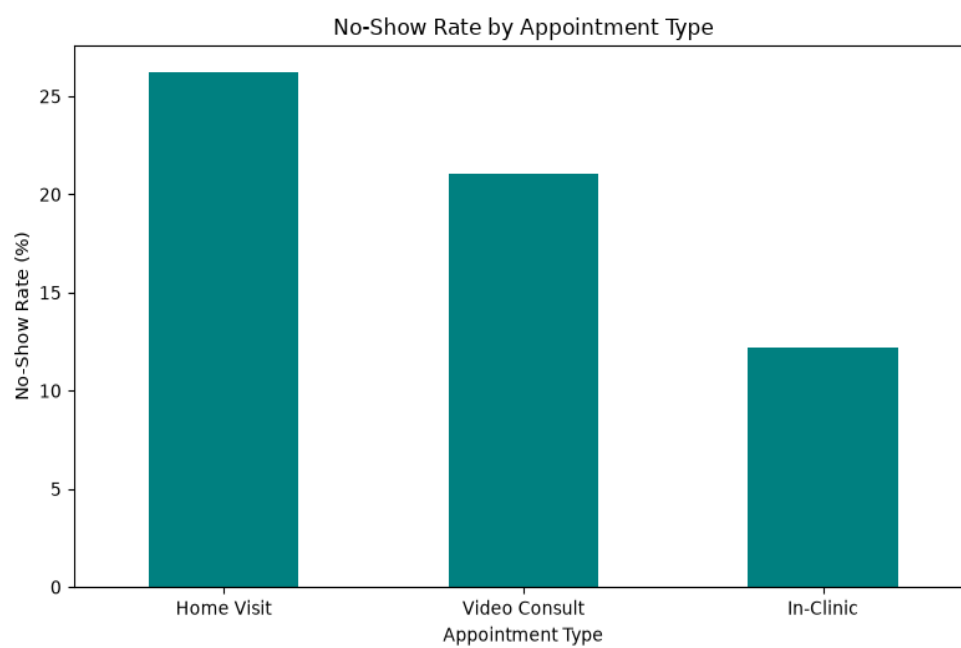


Fig.2.6



Fig.2.7

3. Reminder and Engagement Effectiveness:

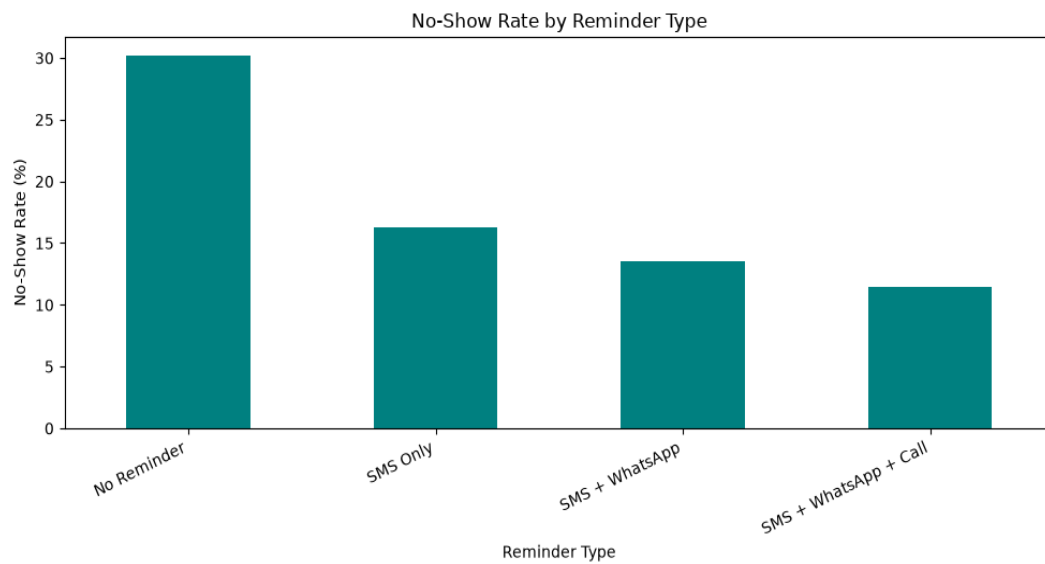


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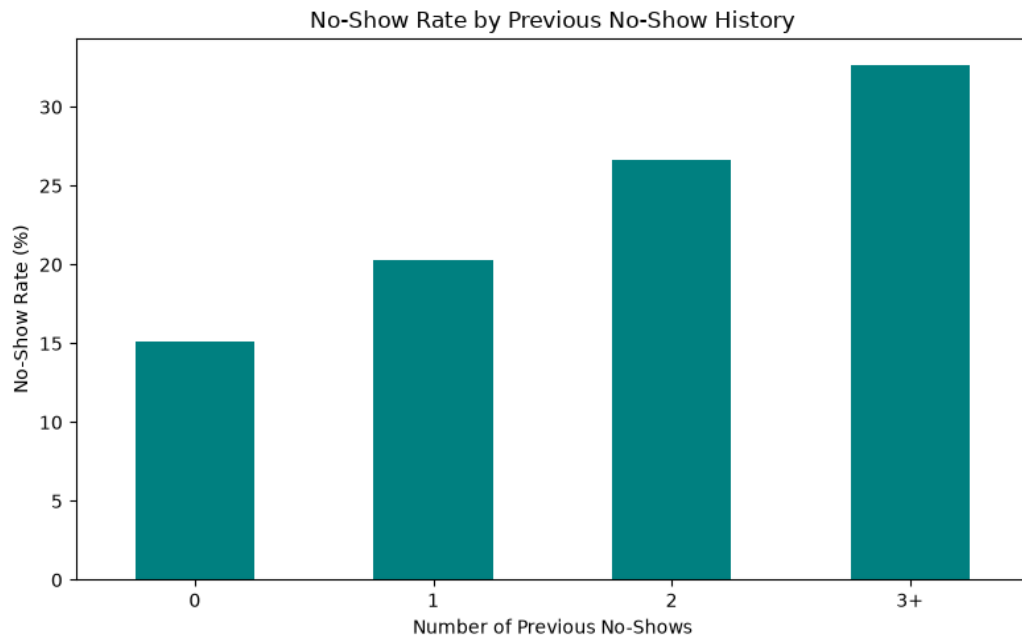


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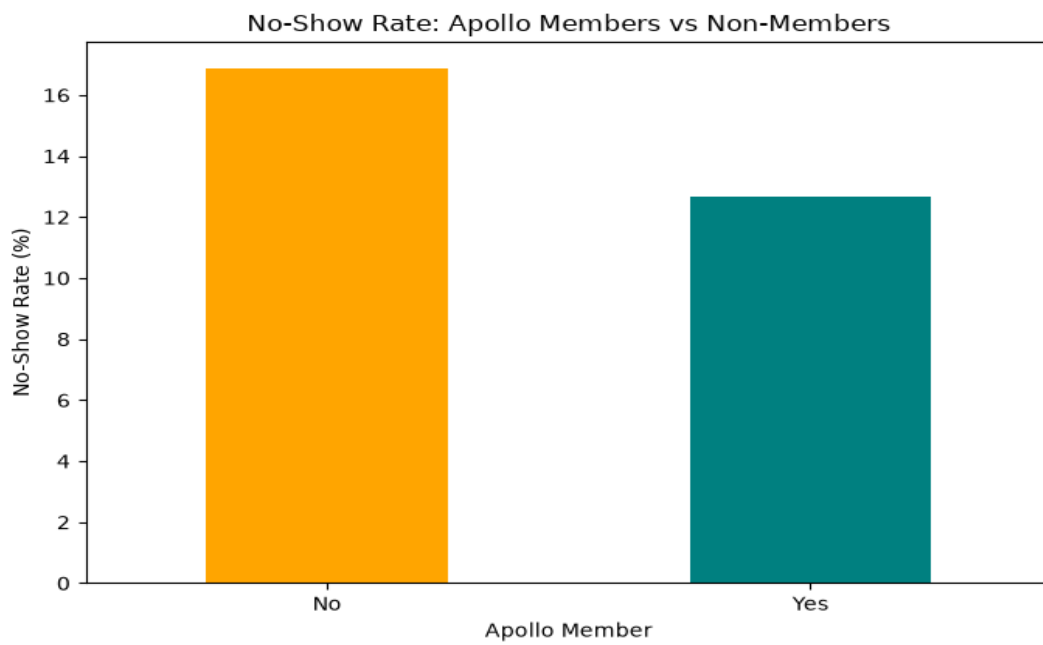


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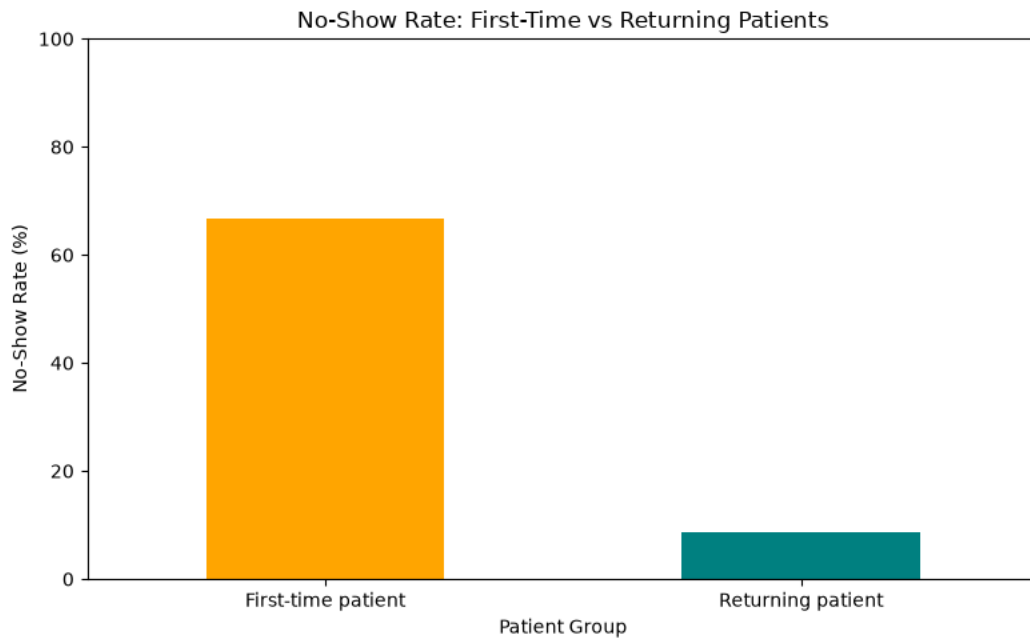


Fig.3.4

4. Patient and Demographic Segmentation:

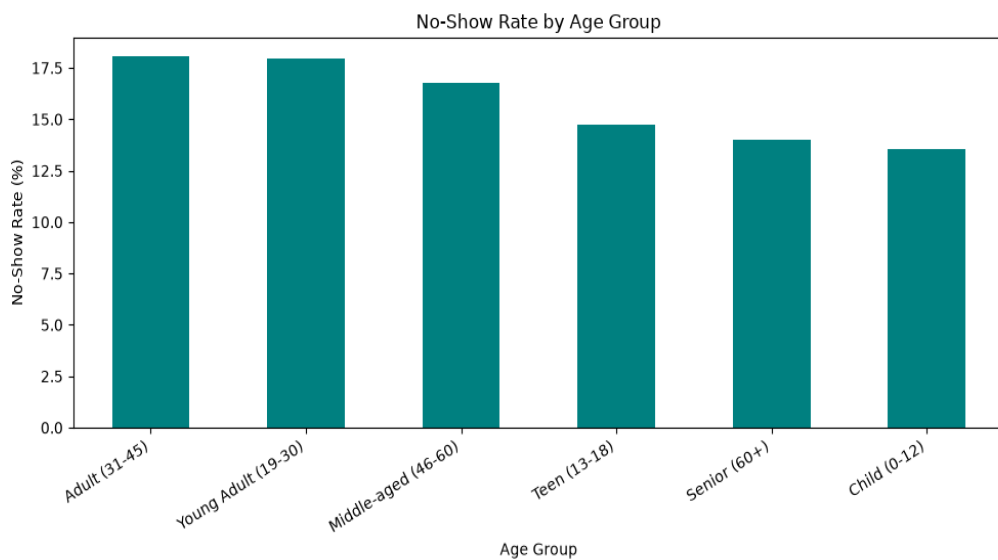


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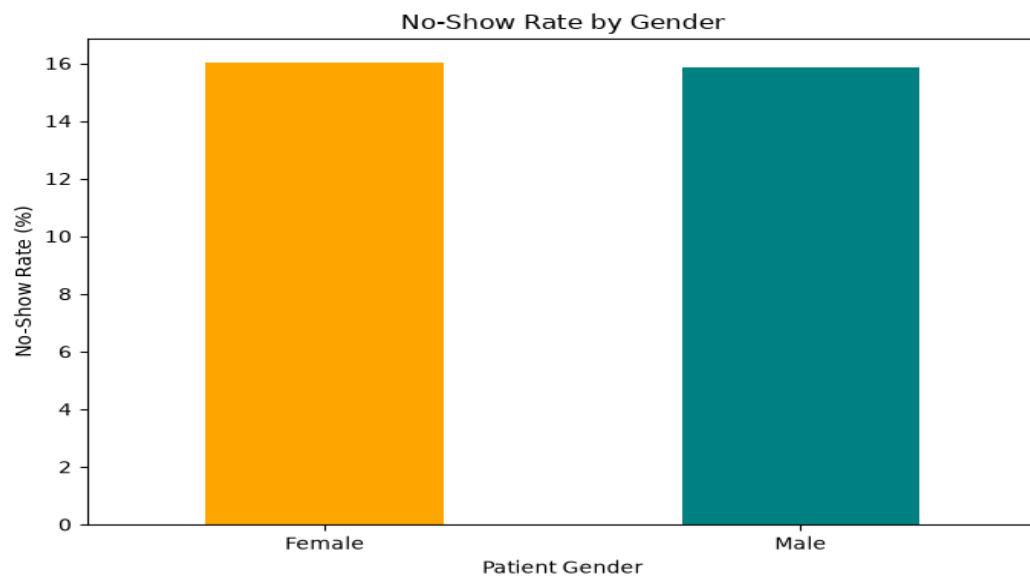


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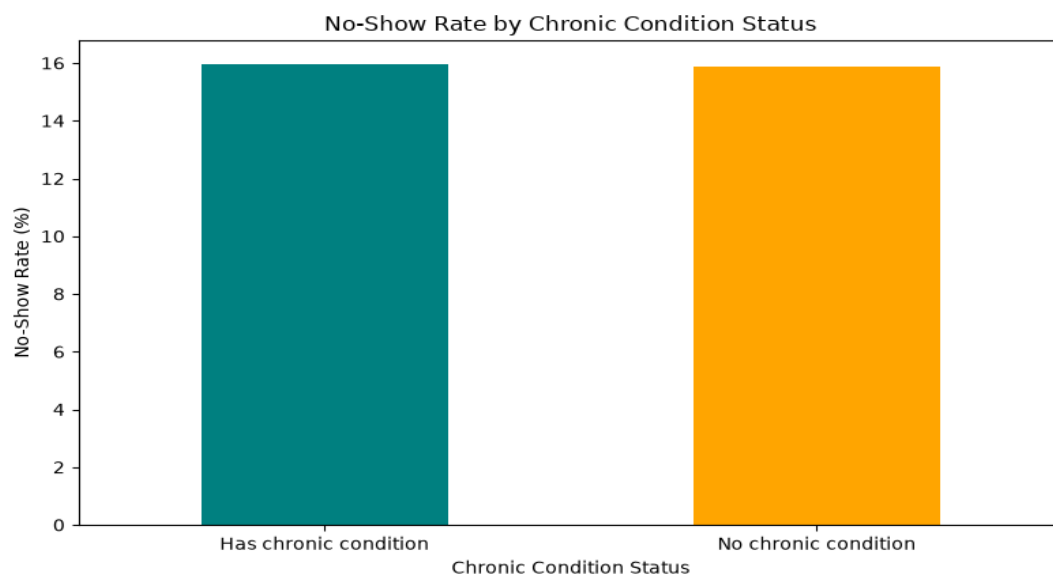


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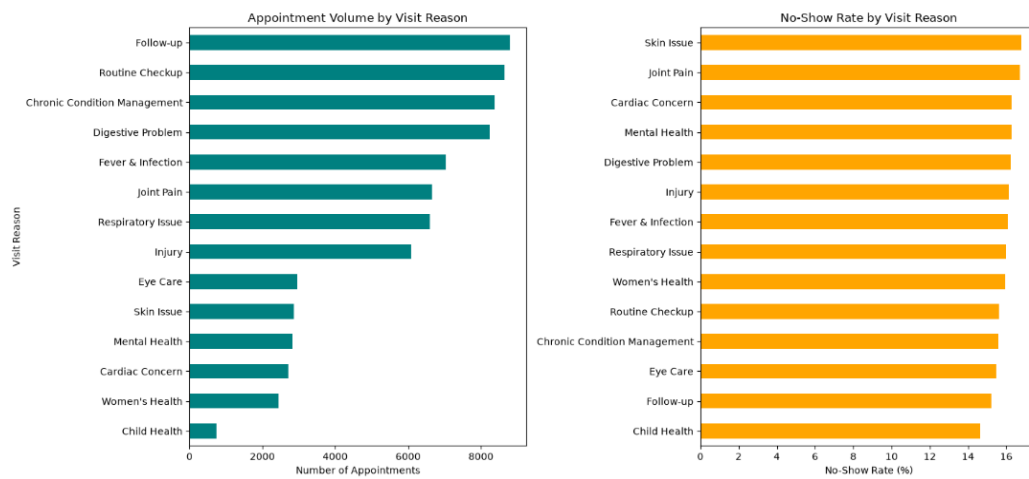


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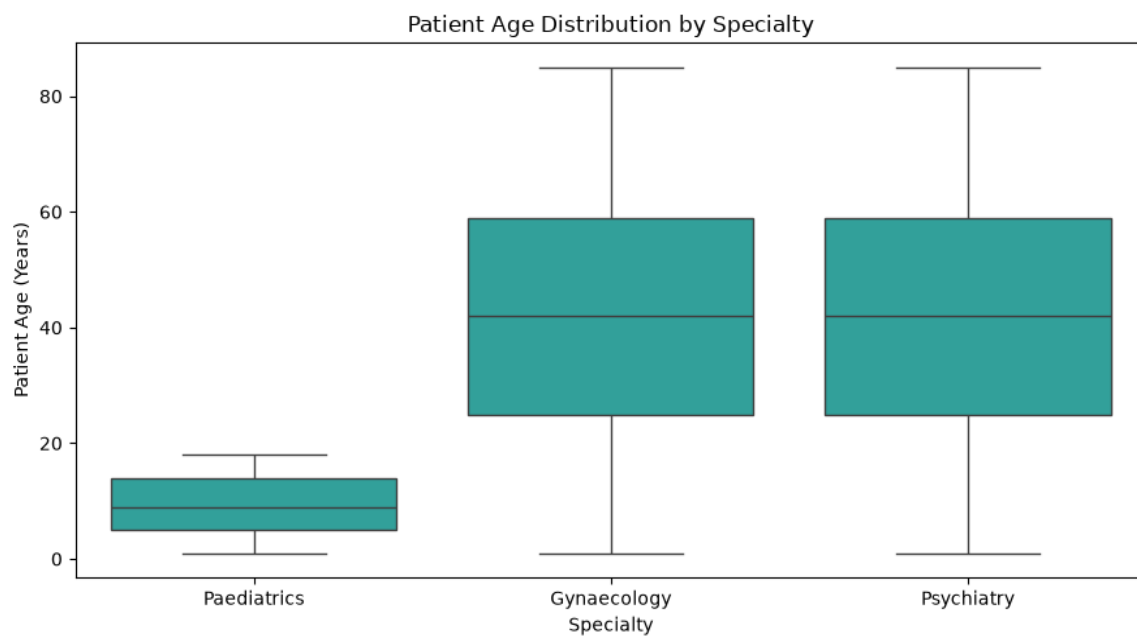


Fig.4.5

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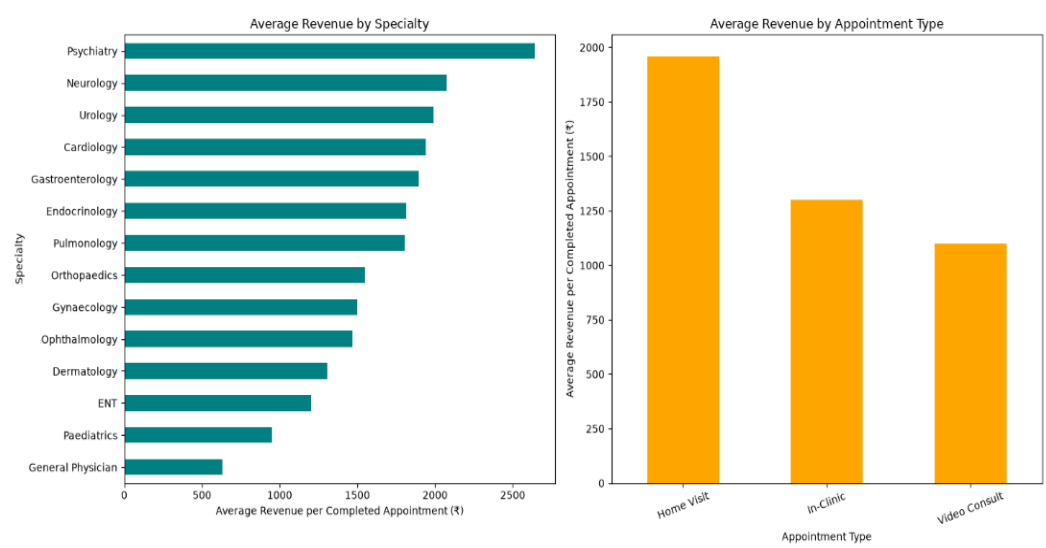


Fig.5.1

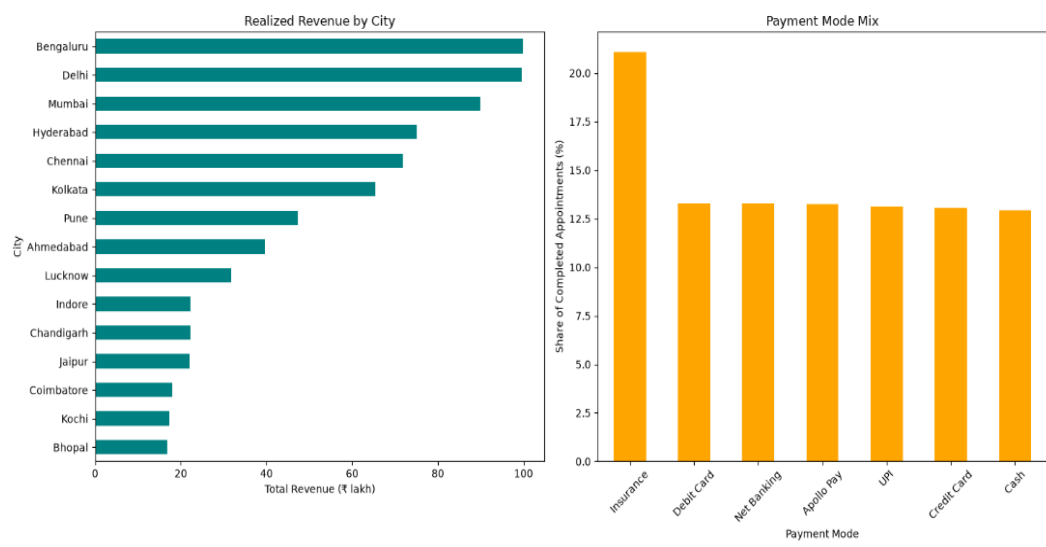


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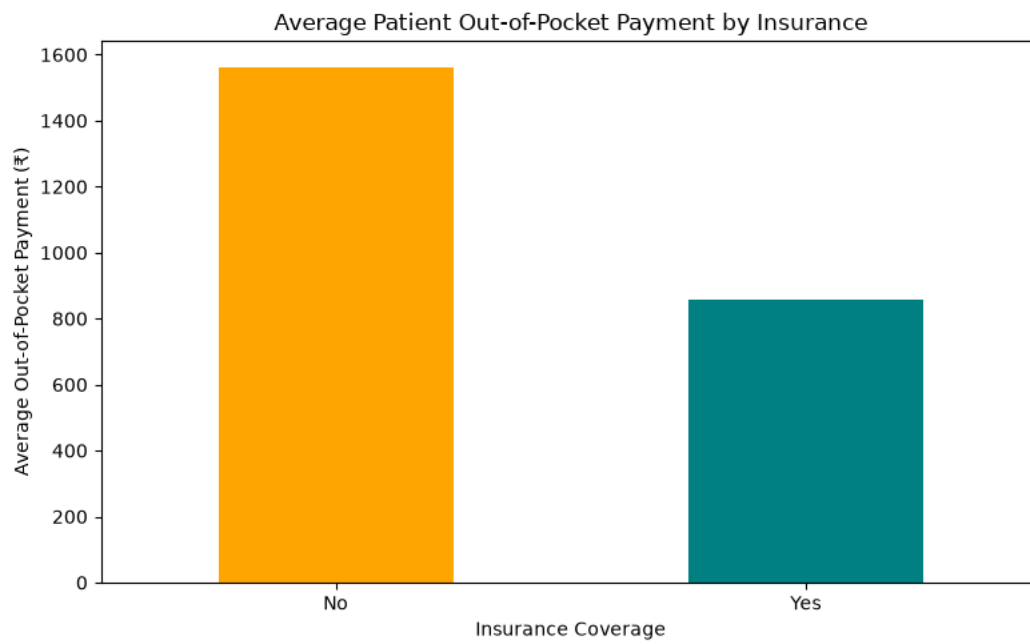


Fig.5.3

6. Doctor Utilisation and Service Quality:

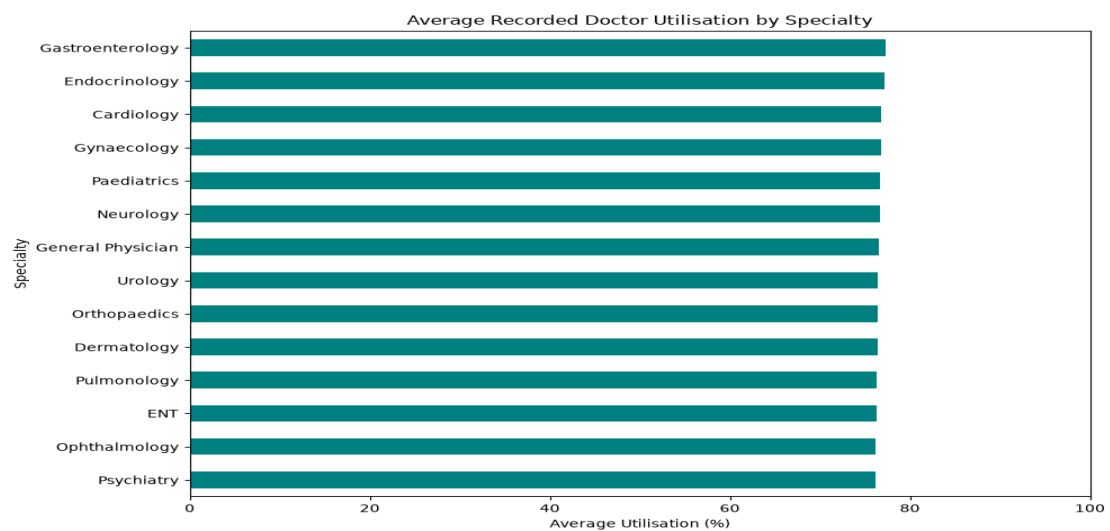


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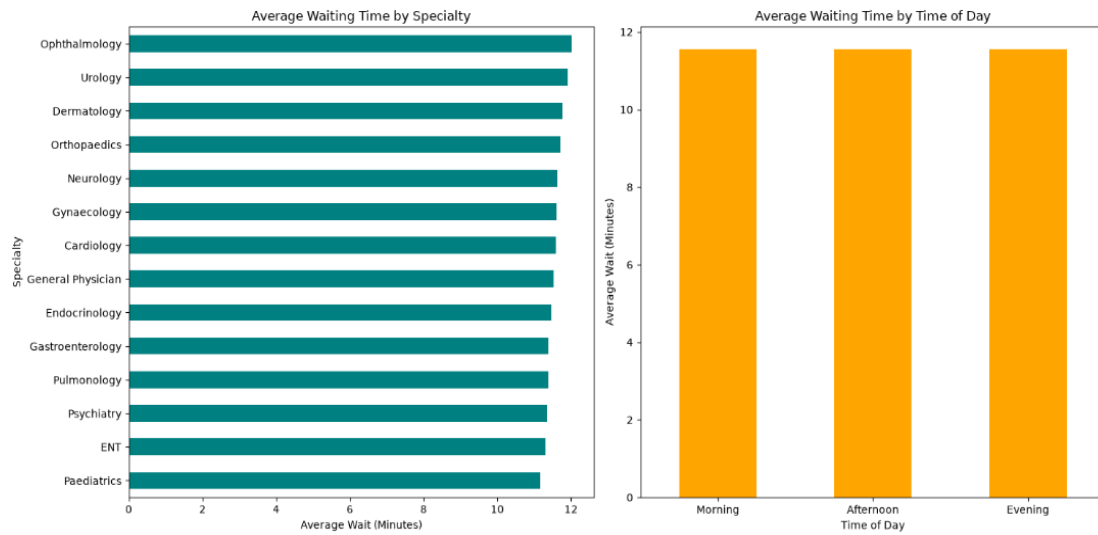


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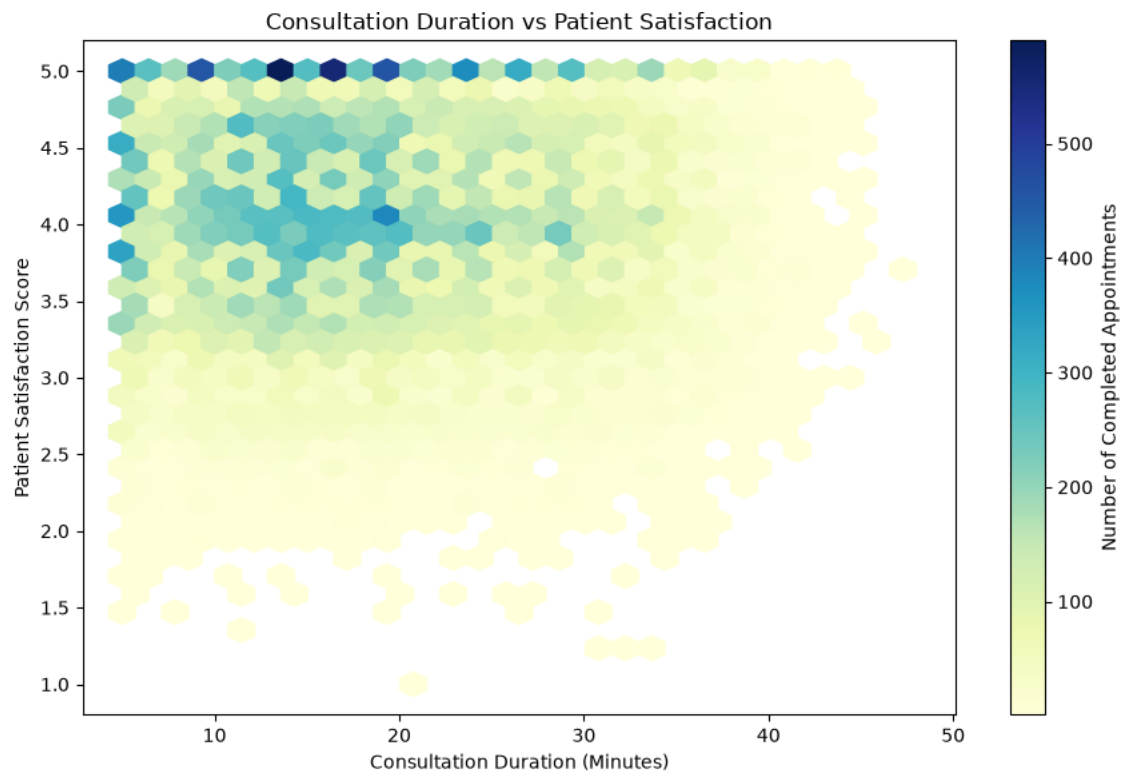


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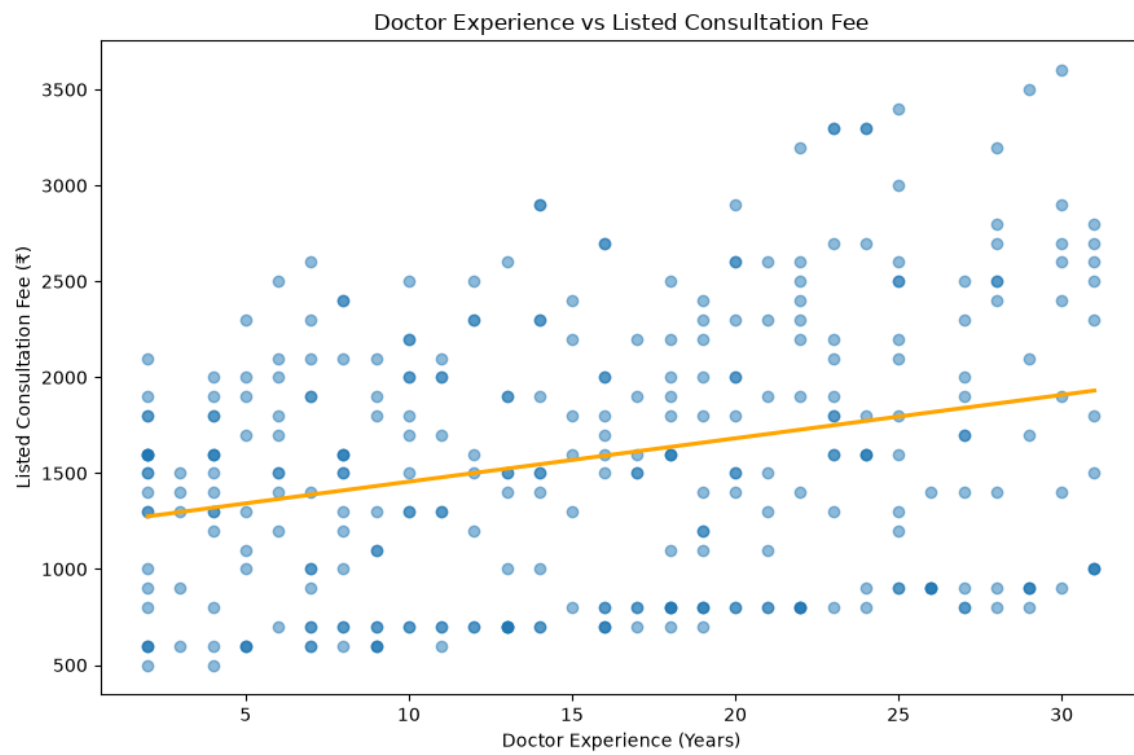


Fig.6.4